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Sex-specific graphs: Relating group-specific topology to demographic and landscape data

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Abstract

Sex-specific genetic structure is a commonly observed pattern among vertebrate species. Facing differential selective pressures, individuals may adopt sex-specific life history traits that ultimately shape genetic variation among populations. Although differential dispersal dynamics are commonly detected in the literature, few studies have used genetic structure to investigate sex-specific functional connectivity. The recent use of graph theoretic approaches in landscape genetics has demonstrated network capacities to describe complex system behaviours where network topology represents genetic interaction among subunits. Here, we partition the overall genetic structure into sex-specific graphs, revealing different male and female dispersal dynamics of a fisher (*Pekania* [*Martes*] *pennanti*) metapopulation in southern Ontario. Our analyses based on network topologies supported the hypothesis of male-biased dispersal. Furthermore, we demonstrated that the effect of the landscape, identified at the population level, could be partitioned among sex-specific strata. We found that female connectivity was negatively correlated with snow depth, whereas connectivity among males was not. Our findings underscore the potential of conducting sex-specific analysis by identifying landscape elements or configuration that differentially promotes or impedes functional connectivity between sexes, revealing processes that may otherwise remain cryptic. We propose that the sex-specific graph approach would be applicable to other vagile species where differential sex-specific processes are expected to occur.

KEYWORDS

genetic network, *Pekania* [*Martes*] *pennanti*, population graph, sex-biased dispersal, sex-specific graphs

1 | INTRODUCTION

Sex-specific genetic structure is a commonly observed pattern among vertebrate species (Lawson Handley & Perrin, 2007; Prugnolle & de Meeûs, 2002). Due to differential selective pressure acting on sexes, sex-specific life history traits may be adopted that ultimately shape genetic variation among populations (Prugnolle, Durand, Theron, Chevillon, & de Meeûs, 2003). Sex-biased dispersal, defined as the propensity of one sex to disperse more than the other, has been recognized as a major process driving male and female difference in genetic structure (Goudet, Perrin, & Waser, 2002; Prugnolle & de Meeûs, 2002). Ultimately, this evolutionary process reduces inbreeding depression and kin competition in wild populations (Dobson, 2013; Greenwood, 1980; Perrin & Mazalov, 2000). There are, however, costs associated with dispersal including competition, unfamiliarity with new sites, difficulty finding mates and increased risks of mortality (Lawson Handley & Perrin, 2007; Prugnolle et al., 2003). Consequently, the direction of sex-biased

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dispersal reflects trade-offs between costs and benefits, which are modulated by sex-specific life history traits, such as mating system and parental investment. For instance, the sex that is most associated with local competition, as intrasexual competition for mates and resources, would be the one more prone to disperse (Greenwood, 1980; Perrin & Mazalov, 2000). Alternatively, the sex that is benefiting from kin cooperation and acquaintance of natal range has reduced pressure for dispersal and is therefore predicted to express philopatry (Prugnolle et al., 2003; Waser & Jones, 1983). Advances in molecular tools have facilitated the indirect detection of sex-biased dispersal patterns (Banks & Peakall, 2012; Favre, Balloux, Goudet, & Perrin, 1997; Fontanillas, Petit, & Perrin, 2004; Goudet et al., 2002) and confirmed its prevalence in numerous taxa, such as mammals and birds, but also its variability in direction and magnitude (Dobson, 2013; Lawson Handley & Perrin, 2007).

Although sex-biased dispersal is commonly reported in the literature, only a few studies have used genetic structure to investigate sex-specific functional connectivity, or how landscape permeability varies between sexes as a result of dispersal behaviour (Proctor, McLellan, Strobeck, & Barclay, 2005; Stow, Sunnucks, Briscoe, & Gardner, 2001). As using telemetry to produce large databases about wildlife dispersal is logistically and economically demanding, a landscape genetics approach may be an effective way to quantify landscape effects on sex-specific dispersal and connectivity (Rioux Paquette, Talbot, Garant, Mainguy, & Pelletier, 2014). Interestingly, studies that have quantified landscape-induced sex-specific responses of dispersal have found that permeability could indeed diverge between demographic groups (Elliot, Cushman, Macdonald, & Loveridge, 2014). In a study that investigated discrepancies in male and female bird dispersal behaviours, Amos et al. (2014) observed that dispersal response to landscape fragmentation and heterogeneity was both species- and sex-specific, affecting to a greater extent the dispersing sex in the Brown treecreeper (*Climacteris picumnus*) and the philopatric sex in the Superb fairy-wren (*Malurus cyaeneus*). In another study, Talbot, Garant, Rioux Paquette, Mainguy, and Pelletier (2012) found that landscape permeability differs between sexes in the striped skunk (*Mephitis mephitis*) and the raccoon (*Procyon lotor*), where female connectivity in both species was negatively affected by highways and major rivers, respectively, and unaffected in males. In a similar resistance-based analysis, Rioux Paquette et al. (2014) detected a positive relationship between genetic distances of female raccoons and the proportion of agricultural fields, but no such relationship existed for males. These studies highlight the importance of investigating how landscape composition and configuration may affect sex-specific genetic structure of vagile species. It underlines the need for interpreting results in the context of species and sex-specific behavioural needs, leading to a better understanding of the selective pressures involved (Anadón, Wiegand, & Giménez, 2012; Lawson Handley & Perrin, 2007).

The recent applications of graph theory in population genetics have demonstrated network capacities to describe complex system behaviours where network topology intuitively represents the genetic interaction among subunits (Dyer, Nason, & Garrick, 2010; Fortuna, Albaladejo, Fernández, Aparicio, & Bascompte, 2009; Garroway, Bowman, Carr, & Wilson, 2008; Kininmonth, van Oppen, & Possingham, 2010; Rozenfeld et al., 2008). The population graph framework decomposes the overall genetic variation into among- and within-strata components, represented parsimoniously as edges and nodes, respectively (Dyer & Nason, 2004). Connectivity in population graphs is determined by conditional independence (Whittaker, 1990), allowing a statistical test of interpopulation covariance (Dyer, 2007). Population graphs also define a measure of conditional genetic distance (cGD) between nodes that has proven to be more statistically robust than traditional pairwise genetic distance measures for recovering underlying demography (Dyer et al., 2010). Despite its apparent utility, population graphs have not been applied to analyses relating sex-specific dispersal processes and covariance structure.

Using biparentally inherited microsatellite markers, the overall goal of this study was to investigate the relative effect of demographic and landscape data on sex-specific network structures of a fisher (*Pekania* [*Martes*] *pennanti*, Erxleben, 1777) metapopulation in the Great Lakes region, Ontario, Canada. A previous network-based study on this metapopulation identified negative relationships between node centrality and snow depth, as well as node centrality and the proportion of immigrants per site (Garroway et al., 2008). Concordant with reciprocating dispersal and density-dependent habitat selection theory (Morris, Diffendorfer, & Lundberg, 2004), these patterns were attributed to a recent fisher population growth and range expansion. These dynamics induced a positive density-dependent dispersal from high-quality to low-quality sites where sites with deep snow, associated with a high proportion of immigrants, were considered to be low habitat quality (Bowman, Donovan, & Rosatte, 2006; Carr, Bowman, & Wilson, 2007). The fisher demonstrates pronounced sexual dimorphism where males are generally 1.5× heavier than females (Powell, Buskirk, & Zielinski, 2003). This species is also known to be intrasexually territorial and polygynous, whereby males often mate with multiple females with no associated parental investment (Arthur, Paragi, & Krohn, 1993; Powell, 1979). Considering the sex-specific ecological characteristics of fishers, we aimed to investigate whether the demographic and landscape effects previously identified at the metapopulation level may be an amalgamation of differential influences exhibited by each sex independently. We had two objectives. First, we tested whether sex-specific demography could lead to a sex-specific network structure. Second, we evaluated the response of sex-specific genetic structure, modelled by network topology, to landscape composition.

Male-biased dispersal has been shown previously in the fisher in the western USA through genetic relatedness analysis (Aubry, Wisely, Raley, & Buskirk, 2004) and telemetry-based studies (Aubry & Raley, 2006; Matthews et al., 2013). We hypothesized that sex-biased dispersal occurs in fishers of the Great Lakes region, and tested this idea using sex-specific graph topologies. In accordance with the fisher mating system, we predicted that males are the dispersing sex. Acknowledging the discrepancies in the literature about evaluating dispersal bias, commonly used tests for detecting sex-

biased dispersal through summary statistics and assignment indices (e.g., Goudet et al., 2002) were used along with our graph approach to investigate potential discrepancies between the two methods.

Snow depth is suggested to be an important factor determining fisher's habitat suitability where soft and deep snow tends to hamper their movement and increase associated energetic cost (Krohn, Elowe, & Boone, 1995; Krohn, Hoving, Harrison, Phillips, & Frost, 2005; Raine, 1983). As juvenile dispersal is realized under winter conditions, the depth of snow might have an important impact on settlement decisions. Considering that this species exhibits a pronounced sexual dimorphism in weight and that males show greater foot loading than females (Krohn et al., 2005), we hypothesized that snow depth has a differential effect on the sexes. We predicted that male connectivity would be more impeded by snow than female connectivity. We used a beta regression approach (Ferrari & Cribari-Neto, 2004) to quantify the relative effect of snow on network connectivity patterns of both sexes.

2 | METHODS

2.1 | Data collection

Data collection occurred from 2000 to 2003, although the majority of the samples (83%) were collected during the trapping season of 2001–2002. Goudet et al. (2002) made several prerequisites for assessing sex-biased dispersal with biparentally inherited markers, including that dispersal occurs at the juvenile stage and that sampling occurs after juveniles have dispersed. The fisher trapping season occurred in fall through mid-winter, while juvenile fishers are generally dispersing (Arthur et al., 1993) prior to the mating period (mating occurs from March to late May) (Powell & Leonard, 1983; Powell et al., 2003). Although age class was not directly assessed, we assumed samples to be representative of dispersal process in fishers. The data set consisted of 722 individuals from 34 landscapes, having an average area of 300 km^2. Landscapes were sampled following a lattice sampling design such that each unit was connected by dispersal, but not daily movement (Arthur et al., 1993; Bowman, Jaeger, & Fahrig, 2002). The majority of samples were collected in southern Ontario, with a few samples collected in neighbouring jurisdictions (i.e., New York and Québec). Landscapes reflected a gradient of snow depth. Weekly mean snow depths from January to April were measured across the area between 1993 and 2002 by the Ontario Ministry of Natural Resources and were thereafter averaged (see Carr, Bowman, & Wilson, 2007 for more details). We subsequently defined each landscape as a node (Figure 1).

Each individual was genotyped at 16 microsatellite loci (see Carr, Bowman, Kyle, et al., 2007; Carr, Bowman, & Wilson, 2007; for further details). The number of sampled individuals in each landscape varied from 3 to 20 for males (9.52 ± 3.95, mean ± *SD*) and 2 to 18 for females (9.27 ± 3.94), corresponding to the identical data set used in previous landscape-based studies published on this species (Carr, Bowman, Kyle, et al., 2007; Carr, Bowman, & Wilson, 2007; Garroway et al., 2008; Garroway, Bowman, & Wilson 2011). Five genetic clusters, previously identified using the Bayesian assignment program STRUCTURE (Pritchard, Stephens, & Donnelly, 2000), composed the actual structure of the metapopulation (Carr, Bowman, Kyle, et al., 2007).

2.2 | Sex-specific graph

We built the sex-specific graph following the population graph method described by Dyer and Nason (2004) using the POPGRAPH package (Dyer, 2014). Similar to how variance is decomposed in the AMOVA framework (Excoffier, Smouse, & Quattro, 1992), population graphs partition the total genetic variation of a system into inter- and intrastrata components represented by connections and node sizes in the graph topology. This results in a saturated network, where all nodes are connected to each other through pairwise genetic relationships. Without any structure assumption, subsequent pruning of the network is achieved based on the strength of pairwise interactions (Dyer & Nason, 2004). In such an approach, each edge is considered as a predictor variable explaining the overall network covariance structure. By the use of the conditional independence approach (Whittaker, 1990), we derived the minimal topological size that described the total observed genetic variance without significant loss of information (Dyer, 2007; Dyer et al., 2010). Edges not pruned were then considered for the calculation of conditional genetic distance (cGD, Dyer et al., 2010), which is the shortest (geodesic) path length separating two nodes. The topological shape of the resulting network is defined by interlocale genetic covariance, which is derived from demographic connectivity (Dyer, 2015; Koen, Bowman, & Wilson, 2016).

2.3 | Sensitivity analyses

The topological structure of a network (i.e., connectivity patterns inside the network) is sensitive to the data involved in its construction. The evaluation of sampling effect on network structure is important considering that un- and undersampled populations might affect network topologies and the resulting model fit when related to landscape data (Naujokaitis-Lewis, Rico, Lovell, Fortin, & Murphy, 2013). However, Naujokaitis-Lewis et al. (2013) did not examine the population graph algorithm specifically, underlying the need for a systematic evaluation of the effect of sampling variability on population graph structure. In another study, Koen, Bowman, Garroway, and Wilson (2013) found that population graph edge lengths were sensitive to sampling intensity, something due to its intrinsic property of estimating covariance structure from all nodes simultaneously. Despite the observed variation in distance estimates, inferences from landscape genetic analysis remained consistent along varying levels of sampling intensities (Koen et al., 2013). Based on these findings, we carried out sensitivity analyses to evaluate sex-specific graph topological variation in both node- and edge-based indices and to assess their consistencies across different scenarios of varying sampling intensity.

Undersampling could particularly be an issue when using sex-specific graphs as the data set is partitioned among demographic

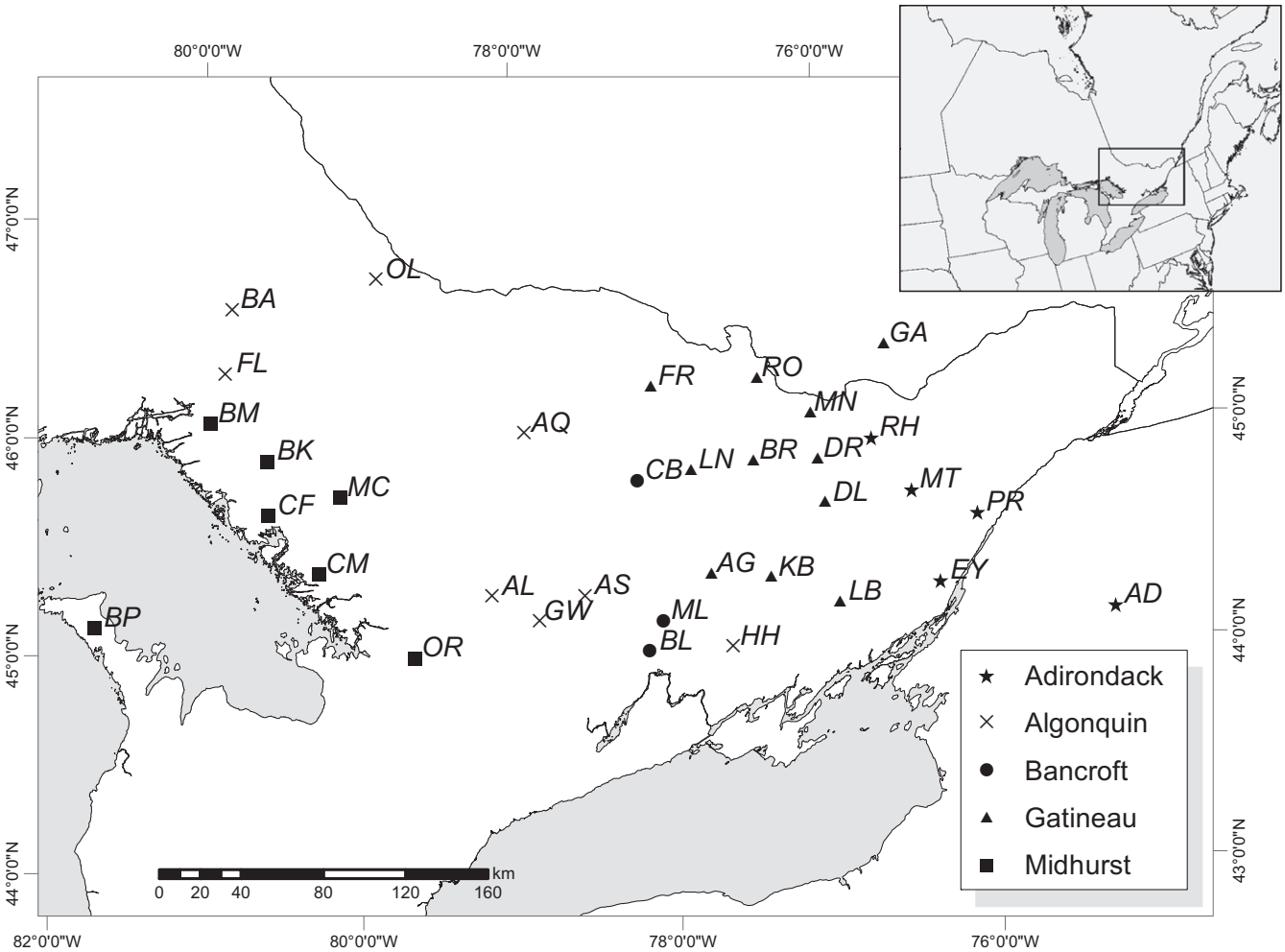

FIGURE 1 Landscape centroid locations in Ontario, Canada. Two letter codes refer to township's IDs, and symbol shapes to sample site assignments to genetic clusters inferred from STRUCTURE modelling

groups. Depending on the sex-specific genetic structure, this reduced number of individuals in each network might potentially affect the genetic variation within each stratum. In order to evaluate this effect, we investigated the minimum number of individuals required per node to obtain a stable estimate of genetic diversity. For that, we conducted a subsampling bootstrap with replacement where genetic diversity was assessed while varying the number of individuals in each node. We chose the effective number of alleles (A_e; Kimura & Crow, 1964) to assess the magnitude of change across a random addition of individuals. We carried out this evaluation on sites that had a minimum number of 11 individuals, represented by 13 landscapes for females and 14 for males. This number reflected the trade-off between having enough scenarios to explore the within-site genetic variation (i.e., starting from 1 individual scenario with a constant increment of one individual up to a total of 11 individuals per site) and having a maximum standardized sample size to reveal the genetic pattern for each sex-specific network. We calculated the genetic diversity (A_e) via the GSTUDIO package (Dyer, 2017) and averaged estimates for each node across replicates. This process was achieved through 999 iterations, and boxplot representations with associated quartiles were thereafter calculated via the GGPLOT2 package (Wickham, 2009).

For facilitating comparisons through scenarios, we calculated the second-order rate of change, an approach commonly used for detecting the optimal number of clusters used in the Bayesian clustering program STRUCTURE (Evanno, Regnaut, & Goudet, 2005; Pritchard et al., 2000). We first calculated the difference in average A_e of each node by $A'_e = A_e n = A_e n - 1$, where n corresponded to the bootstrap scenario involved (i.e., a minimal number of individual in each node). We then computed the absolute variation in the parameter among the consecutive scenarios with $|A''_e| = |A'_e n + 1 - A'_e n|$. To apply a conservative approach, we selected the first absolute value of the second-order rate of change that described a constant rhythm of change through the different bootstrapping scenarios.

As the fisher has a capacity for extensive movement (Roy, 1991) and may colonize in a stepping-stone fashion (Wisely, Buskirk, Russell, Aubry, & Zielinski, 2004), connectivity might be the result of both proximate and more distant subpopulations (Watson et al., 2011). Indeed, using the same data set, the combination of both direct and indirect network connections has been shown to be more

related to demographic and landscape data compared to the use of direct connections solely (e.g., degree centrality) (Garroway et al., 2008). Eigenvector centrality (EVC; Bonacich, 1972) is a topological measure that involves both local (i.e., how well node is connected, also named degree centrality) and regional (i.e., how well local neighbours are connected) centrality. The robustness of eigenvector centrality measurements to missing data (e.g., node) has, however, been shown to depend on sampling design and intrinsic network properties (Borgatti, Carley, & Krackhardt, 2006; Costenbader & Valente, 2003; Herland, Pastran, & Zhu, 2013). To evaluate its robustness to node removal under the population graph framework, topological sensitivity was determined by monitoring the magnitude of change in centralization while randomly removing an incremental number of nodes (i.e., unsampled sites). The eigenvector centralization score (CS), calculated as: $\mathrm{CS} = \sum_{i=1}^{n}(\max(\mathrm{EVC}) - \mathrm{EVC}_i)$ where n corresponds to the number of nodes in a specified graph (Wasserman & Faust, 1994), is a network-level index based on node centrality ranging between 0 and 1, where 1 corresponds to the most centralized network (Freeman, 1979). The choice of this index was motivated by its property to be formerly standardized, letting the opportunity for direct comparison across networks (Freeman, 1979). In each scenario, we executed a random removal selection of nodes based on the specific scenario tested (i.e., number of populations removed) and calculated the centralization score. All results were kept under the Euclidean norm and standardized by dividing the corresponding centralization score by the maximum theoretical score for a network of the same size (i.e., same number of nodes) (Ruhnau, 2000; Wasserman & Faust, 1994). Following the same random removal selection approach, we conducted an edge-based analysis by monitoring the averaged shortest path length, or conditional genetic distance (cGD). All edge weights were conserved for all topological metrics. Disconnected nodes were discarded from the shortest path length calculation, avoiding to attribute arbitrary weight to nonexistent edge. We used the IGRAPH package (Csárdi & Nepusz, 2006) for the eigenvector computation and the GSTUDIO package (Dyer, 2017) for the calculation of cGD.

We used an identical bootstrapping procedure for both male and female graphs where each scenario was iterated 999 times. We evaluated the effect of node removal on topological metrics by the magnitudes of change observed across scenarios. All algorithms were written in R (R Development Core Team, 2014) and are available at https://github.com/PhilBertrand/SexSpecificGraph.

2.4 | Sex-biased dispersal in fisher

We investigated the male-biased dispersal hypothesis using five different statistical tests conducted in FSTAT version 2.9.3.2 (Goudet et al., 2002). All tests are based on summary statistics and thus rely on structuring and clustering assumptions (Pearse & Crandall, 2004). To investigate potential discrepancies between the clustering approaches and levels of analysis, each test was carried out using two grouping schemes: the first used the sampled nodes as grouping entity and the second used genetic clusters based on assignments from the Bayesian clustering program STRUCTURE (Carr, Bowman, Kyle, et al., 2007) (Figure 1). Population structure parameters and assignment index (Favre et al., 1997; Paetkau, Calvert, Stirling, & Strobeck, 1995) were used to assess whether one sex dispersed more than the other. Based on the rationale that dispersal increased gene flow, lowering the allelic variance among groups compared, we expected F_{ST} (Weir & Cockerham, 1984) and the relatedness index (r) to be greater in the philopatric sex (i.e., hypothesized as female in this study). If the migrant and nonmigrant ratio differs between sexes, then a relative heterozygote deficit due to the Wahlund effect is expected in the dispersing sex. F_{IS} was thus predicted to be greater and positive in male fishers.

Finally, we computed the mean (mAIc) and the variance (vAIc) of the corrected assignment index (AIc) for genetic diversity (Favre et al., 1997; Paetkau et al., 1995). The index calculated the probability that a given genotype originated from the site where it had been collected. In a sex-biased dispersal context, the dispersing sex should have a lower mean (i.e., the lesser probability of originating from the sampling site) and a greater variance considering that not all members of the same sex disperse. Significance of all tests was based on a randomization procedure where, in each iteration, individual genotypes in each subgroup were randomly assigned a sex while keeping the original sex ratio. Observed and randomized statistics were then calculated by (1) the observed difference between male and female F_{ST}, F_{IS} and r, (2) the ratio of largest vAIc over the smallest one and (3) the t-statistic involving mAIc, generating random distributions with the null hypothesis that males and females disperse equally. The probability supporting the null hypothesis was then calculated as the proportion of time the randomized value was greater or equal to the observed one (see Goudet et al., 2002 for a more detailed description). These various statistics are known to have varying levels of sensitivity in detecting bias in dispersal rate where vAIc is more sensitive to low and F_{ST} to high dispersal rate, while mAIc is in between (Goudet et al., 2002). Male-biased dispersal has been previously described in the fisher in the western USA (Aubry & Raley, 2006; Aubry et al., 2004; Matthews et al., 2013) but remains unobserved in some part of the species' range (Arthur et al., 1993). Based on this prior knowledge, we conducted two-tailed tests testing the sex-biased dispersal hypothesis with 9,999 iterations. The two-tailed test is known to have reduced power compared to the one-tailed test (Goudet et al., 2002), and the choice of this former is seen as more rigorous considering the ambiguity regarding the sex-biased dispersal magnitude in the fisher.

2.5 | Network properties & Intranodal analysis

We used a two-tailed Mann–Whitney U test to compare eigenvector centrality distribution between male and female graphs. Due to unequal network size, comparison was made on the eigenvector centrality-normalized distributions, where centrality scores were multiplied by $\sqrt{2}$ (i.e., Euclidean norm; Ruhnau, 2000). Based on the rationale that sex-biased dispersal might generate differences in male and female graph topologies, we hypothesized that centrality

distributions should be different between the two sex-specific graphs.

By using regression approach, Garroway et al. (2008) found different relationships associating centrality measurements from the entire population graph (i.e., males and females together) with the landscape proportion of immigrants and snow depth. By partitioning the entire graph into two unisex networks, we repeated these analyses to consider the sex-specific component in landscape analysis.

Following Carr, Bowman, & Wilson (2007), we used the proportion of immigrants identified through the Bayesian clustering program STRUCTURE (Falush, Stephens, & Pritchard, 2003; Pritchard et al., 2000). In this case, migrants, or effective migrants having bred as reflected in the ancestry coefficient, were identified by their respective ancestries ($q_1 \geq 0.6$) as belonging to a different cluster than the one where they had been sampled (Figure 1; Carr, Bowman, & Wilson, 2007). We used the proportion of immigrants as an index of connectivity among sites, and combined sexes in the analysis for this purpose.

Eigenvector centrality distribution is, however, commonly skewed to the right and bounded, lying theoretically between [0,0.707] but commonly scaled as we have done to [0,1], facilitating the interpretation of centrality estimates (Newman, 2010; Wei, Pfeffer, Reminga, & Carley, 2011). As sex-specific topologies are tested independently, scaling was achieved using the maximum norm, consisting in dividing centrality scores by the maximum score of the centrality vector (Ruhnau, 2000). Fitting ordinary least-squares regression to bounded and skewed response variable shows, however, the risk of violation of the linearity, normality and homoscedasticity assumptions (Cribari-Neto & Zeileis, 2010; Schmid et al., 2013). Although transformation of the response distribution (e.g., logit or arcsine functions) is commonly used for complying with model assumptions, the resulting estimates might have biased predictive capacity and may be difficult to interpret (Cribari-Neto & Zeileis, 2010; Warton & Hui, 2011).

One alternative to data transformation is the use of the beta regression model, which assumes the response variable to be constrained to the unit interval and to follow the beta distribution (Cribari-Neto & Zeileis, 2010; Ferrari & Cribari-Neto, 2004). The beta density is parameterized by both a mean (μ; location) and a dispersion (ϕ; precision) parameters (Ferrari & Cribari-Neto, 2004). These properties make the beta distribution flexible for regression using variable from the unit interval, which typically shows nonconstant variance with less variation at their limit boundaries (Cribari-Neto & Zeileis, 2010). Acknowledging that the models might not have a constant dispersion, link functions are used for relating the observations to linear predictors of both the mean and dispersion submodels (Cribari-Neto & Zeileis, 2010).

Considering the intrinsic nature of the response variable distributions, we conducted beta regressions with the BETAREG package (Cribari-Neto & Zeileis, 2010) to investigate the relationship between node centrality of each sex-specific graph (i.e., dependent variable) over the proportion of immigrants and the snow depth (i.e., independent variables). Because the model assumes that values of the response variable (y) lie strictly on the unit interval ($0 < y < 1$), we fitted limit boundaries of y by $(y \cdot (n - 1) + 0.5)/n$, where n corresponds to the sample size (Schmid et al., 2013). We also controlled for potential random variation in sampling intensities by adding the number of individual males and females sampled within each node as regressor of their respective sex-specific model. According to a previous study performed on the same data set (Carr, Bowman, & Wilson, 2007) and the correlation level between the sample size of males and the snow depth (males: $R^2 = .108$, $p = .066$; females: $R^2 = < .001$, $p = .959$), we first tested the interactions effects in full models and in a second step, tested main effects based on the corresponding hypothesis (i.e., demographic or landscape). Considering the intimate relationships among covariates, all predictors were first centered to their mean to avoid multicollinearity (Robinson & Schumacker, 2009), generating covariates having a mean 0 while keeping their original standard deviation. A variance inflation factor (VIF) of <2 occurred for all covariates in the two full sex-specific models after centering indicating uncorrelated predictors (Dormann et al., 2013). We selected adequate parameterization of the dispersion submodels based on the predictors' ability in modelling heteroscedasticity and confirmed their contribution significance by likelihood ratio tests for nested model (significant threshold of $p \leq .05$), comparing model fits with and without the addition of the corresponding predictors (Bursac, Gauss, Williams, & Hosmer, 2008; Ferrari & Cribari-Neto, 2004). In order to avoid spurious results due to nonindependence of centrality-based data (Farine & Whitehead, 2015), we evaluated significance of predictor estimates by two-tailed permutation tests, along 9,999 iterations. In each iteration, we permuted node centralities and refitted the beta model, thus generating a random distribution of each coefficient. We considered an effect to be significant if the observed coefficients were greater or less than 95% of the generated (absolute) estimates.

We tested whether models were deviating from the null one using again the likelihood ratio test based on the nominal value of 0.05. Link functions for modelling both location and precision parameters were chosen in function to comply to model postulate and maximize goodness of fit (Ferrari & Cribari-Neto, 2004). In all cases, we used the bias reduction approach for the computation of the maximum-likelihood estimators (Grün, Kosmidis, & Zeileis, 2012). We finally assessed the model's predictive power using the generalized determination coefficient (R^2_{pp}) from Zheng and Agresti (2000), measuring the squared correlation between the observed and predicted values derived from the models (Baguley, 2012; Zheng & Agresti, 2000).

3 | RESULTS

3.1 | Sensitivity analyses

Monitoring of the effective number of alleles (A_e) through the incremental addition of one individual per node indicated a stabilization of variation at approximately 5–6 individuals (Figure 2a,b). Although similar, bootstrap distributions for each scenario indicated a larger variation in genetic diversity (Figure 2a) and a more regular pace of

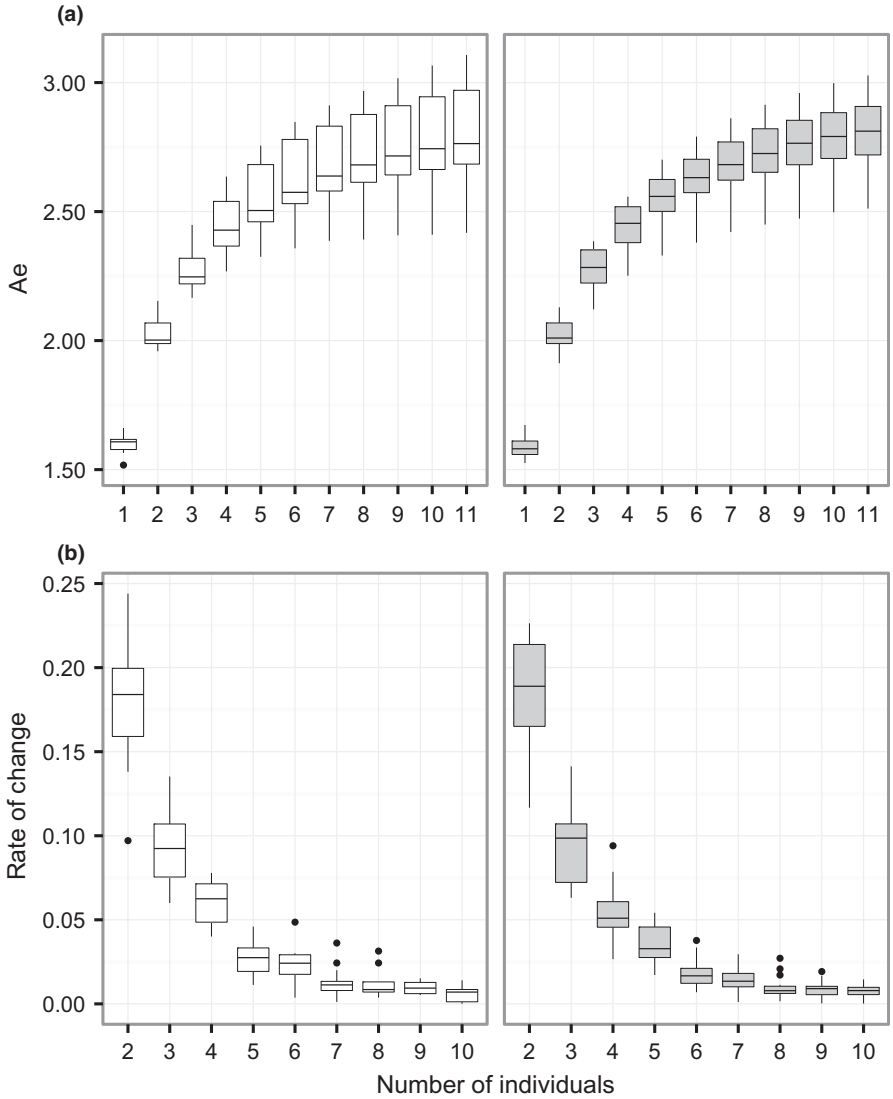

FIGURE 2 Parameter distributions (median, 25th and 75th quartiles) for each bootstrap scenario. (a) Results from the mean effective number of alleles (A_e) of every node and (b) their second-order rate of changes when varying the minimal number of individuals per node of female (white) and male (grey) networks, after 999 iterations

change for females than for males across nodes (Figure 2b). To be conservative, we selected the number where the variation in the second-order rate of change of the whole distribution was below a threshold of 0.05, leading to a selection of six individuals per sex as a minimum number of genotypes within each node of male and female graphs. All subsequent analyses were based on this minimal value that resulted in a reduction in the number of landscapes from 34 to 27 and 28 for male and female graphs, respectively. After this reduction, the number of sampled individuals in each landscape varied from 6 to 20 for males (10.70 ± 3.32, mean ± *SD*) and 6 to 18 for females (10.32 ± 3.27).

We evaluated thereafter topological sensitivity via a random, sequential removal of 1–24 nodes, corresponding to a scenario where networks are completely collapsed (Figure 3). Although females demonstrated more important variability within scenarios, magnitudes of score changes were consistent between male and female networks. Both networks adopted a slight and constant increase in centralization up to the removal of the 9th for males and 13th for females, representing a difference in the averaged centralization scores from the first node removal of 0.114 for males and 0.156 for females, respectively. Mean pace of monotonic changes across these scenarios were, respectively, of 0.014 for males and 0.013 for females.

Following a similar approach to the previous analysis, the subsequent random removal of 1–24 nodes showed a general constant and slight increase in shortest path lengths up to the 13th node for both males and females (Figure 4). Averaged trend in males was, however, nonmonotonic, showing a slight decrease at scenario 8. Variation both within and among scenarios increased subsequently to stabilize around the removal of the 19th node. Difference in averaged shortest path length from the removal of the first node up to the 13th was of 7.885 for males and 7.750 for females.

3.2 | Sex-biased dispersal in fisher

In order to allow for direct comparison among landscapes and ensuring a minimum number of individuals of both sexes in the randomization process (Goudet et al., 2002), we performed all tests on the

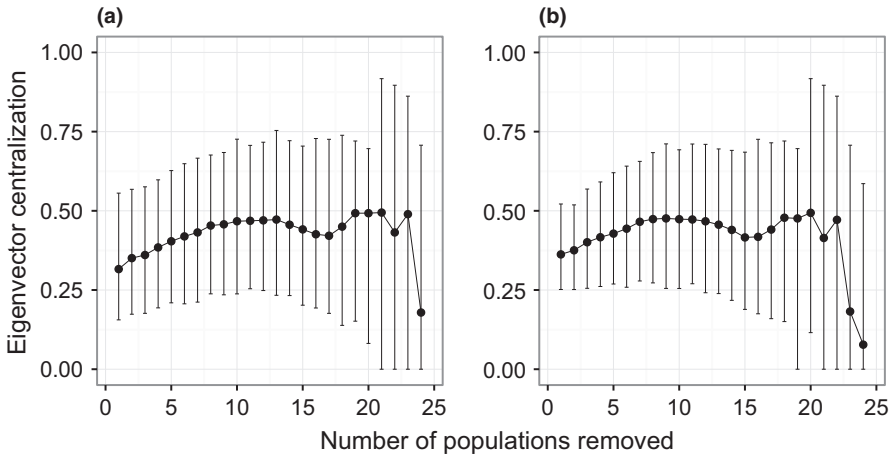

FIGURE 3 Topological variation monitored via normalized centralization score (95% CIs) for each bootstrap scenario. Results for female (a) and the male network (b) when varying the number of populations removed per network, after 999 iterations

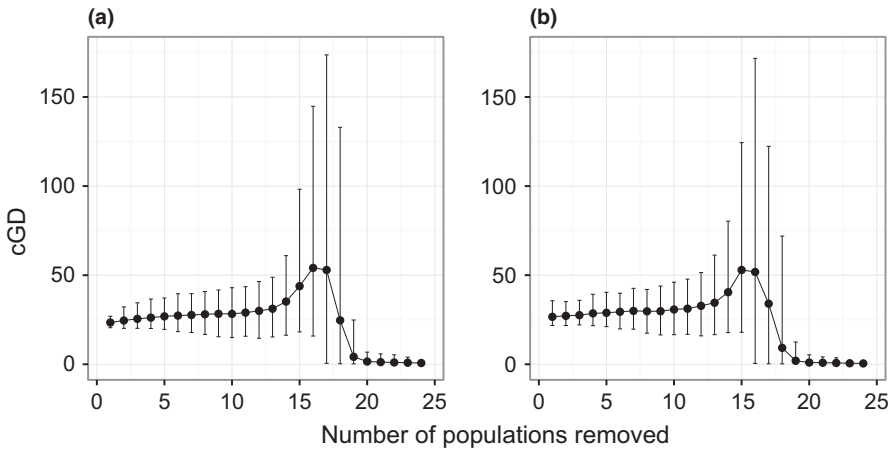

FIGURE 4 Topological variation in network-level averaged shortest path length (95% CIs), or cGD, for each bootstrap scenario. Results for female (a) and the male network (b) when varying the number of populations removed per network, after 999 iterations

standardized sample of 25 landscapes, corresponding to a minimum sample size of six males and six females. Congruent results were found between the two selected sampling schemes, where two of five tests indicated a significant difference between sexes and supported the male-biased geneflow hypothesis (Table 1).

3.3 | Network properties & Intranodal analysis

Topological measurements were computed from both male and female sex-specific graphs (Figure 5). Although showing the expected trend, the eigenvector centrality-scaled distribution showed no significant difference between males and females ($\bar{x}_M \pm SD = 0.376 \pm 0.425$; $\bar{x}_F \pm SD = 0.322 \pm 0.337$; $U = 368.5$; $p = .880$).

Full beta regression models confirmed that predictor interactions had no significant effect on male and female centrality (see Table S2). In consideration of prior hypotheses, main effects of both demographic and landscape predictors were thereafter investigated. The final demographic male model deviated from the null one ($\chi^2(3) = 11.069$; $p = .011$), where the proportion of immigrants showed a significant and inverse relationship to node centralities (Table 2). In order to confirm this parameterization, we conducted a backward deletion test for the mean (μ) submodel (likelihood ratio test; nominal level <0.05) with fixed dispersion (Bayer & Cribari-Neto, 2017) and then, confirmed dispersion (ϕ) submodel parameters based on selected location predictors (see details in Table S3). This procedure confirmed that the remaining predictors poorly accounted for the variability of the observed data (Table S3). Parameterizing the dispersion submodel with the proportion of immigrants did not improve model's fit ($\chi^2(1) = 3.417$; $p = .065$), but showed an improvement with the addition of the number of sampled males ($\chi^2(1) = 4.998$; $p = .025$). Although its inclusion was supported by the likelihood ratio test, permutation procedure revealed that the sampling variable had no significant effects in both submodels (Table 2). The slope estimate from the final demographic female model was, moreover, not different from zero (Table 2). As a result, the final female regression did not diverge from the null model ($\chi^2(2) = 0.597$; $p = .742$) and adding the proportion of immigrants ($\chi^2(1) = 3.311$; $p = .577$) and the number of sampled females ($\chi^2(1) = 1.045$; $p = .307$) as predictors of the dispersion submodel did not improve the model's fit.

In consideration of the landscape-based hypothesis, the main effect of snow depth on male and female centrality was subsequently evaluated. Although being neutral to male centrality, resulting in a nonsignificant final model ($\chi^2(1) = 0.440$; $p = .507$) with no improvement when adding the predictor as a precision parameter ($\chi^2(1) = 1.028$; $p = .311$), snow depth showed a significant and negative effect on female centrality (Table 3), while explaining a large

TABLE 1 Two-tailed tests based on 9,999 randomizations[a] of three genetic parameters and two assignment indices using two sampling schemes, where the first considered the nodes as sample groups (n = 25) and the second the genetic clusters inferred from STRUCTURE analysis (Carr, Bowman, Kyle, et al., 2007)

	N	M		F		p^b	
n	526	262		264		–	
		Node	Cluster	Node	Cluster	Node	Cluster
F_{ST}	0.069	0.061	0.045	0.081	0.061	.007	.017
F_{IS}	0.070	0.076	0.098	0.061	0.090	.412	.638
r	0.122	0.108	0.080	0.142	0.107	.007	.017
mAIc	–	−0.256	−0.304	0.254	0.301	.148	.143
vAIc	–	15.792	23.416	15.840	20.965	.983	.490

[a]Calculated from FSTAT (Goudet et al., 2002).
[b]Testing the difference between males (M) and females (F).

part of its variation (R^2_{pp} = ~0.70) (Figure 6). The use of a backward deletion test (likelihood ratio test; significant threshold of 0.05) supported the selected parameterization of the final landscape female model (see details in Table S3), which was highly significant ($\chi^2(1) = 22.165$; $p = <.001$) with no improvement associated with the addition of the snow depth as a predictor of the dispersion submodel ($\chi^2(1) = 1.607$; $p = .205$).

4 | DISCUSSION

We aimed to understand the implication of considering intrasexual genetic structure, modelled as a network, to investigate the sex-specific relationship to demography and landscape composition. Representing the fisher metapopulation as a network gave the opportunity to use network topologies as population genetic metrics. Our findings were consistent with the general hypothesis that metapopulation-level landscape effects could be partitioned between sexes. Furthermore, the apparent ability of the proposed approach for detecting sex-biased dispersal suggests that sex-specific networks are indeed sensitive to male and female demographic processes like dispersal.

As inferences are constrained by the data used in the construction of networks, it is recommended to do topological sensitivity analyses to evaluate the resulting estimates (Koen et al., 2013; Naujokaitis-Lewis et al., 2013). Calls for investigating sampling design effects on landscape genetic analysis conclusions have been stressed many times (Balkenhol et al., 2009; Hale, Burg, & Steeves, 2012; Landguth et al., 2012). Although data resolution for diversity estimates is increasing with codominant single-nucleotide polymorphism availability, microsatellite-based studies are still widely used in wildlife genetic research (Queirós et al., 2015; Storfer, Murphy, Spear, Holderegger, & Waits, 2010). From a site-based perspective and using microsatellite genetic markers, Hale et al. (2012) found across four taxa that the accuracy of diversity estimates stabilized at a sample size of 25–30 individuals per site. Using networks, Koen et al. (2013) found congruent results where the accuracy of graph distance estimates (or cGD) increased with both the number of sample sites and individuals per site, but that fewer samples were needed for stronger genetic structure. However, when relating cGD to landscape features, more statistical power was gained by maximizing the number of sampling sites over the number of individuals per site (Koen et al., 2013). Population graphs are particularly sensitive to such questions, because network covariance structure is generated concurrently from all genetic relationships among the investigated nodes (Dyer & Nason, 2004).

Finding optimality between node number and the minimal number of individuals within a node depends largely on the underlying genetic structure of the species, which in turn relies on dispersal dynamics (Prugnolle et al., 2003). Considering the potential differential selective pressure on both sexes, optimality might diverge across sex-specific graphs. Obviously, performing sex-specific analysis needs division of the original data set into demographic groups, having potential impacts on the statistical power and inference reliability. In this project, we aimed to identify the optimum between the minimal number of individual per site that stabilizes the genetic diversity estimate variations within the node and the number of nodes that induce a considerable change in the topologies. We found that both node- and edge-based indices of fisher networks were resilient to the low level of node removal. Analysis based on centralization scores showed that both male and female graphs adopted first a monotonic increase in centrality, indicating an increasing heterogeneity among node connections before stabilizing at a moderate centralization level. Congruent with this result, the sensitivity analysis conducted on the averaged shortest path length demonstrated a great robustness of both networks from low to moderate level of node removal, indicating that although network centralization increases across scenarios, shortest path length among nodes remains relatively constant, likely due to redundancy in alternative paths among nodes (Albert, Fortuna, Godoy, & Bascompte, 2013; Garroway et al., 2008). These results suggest that nodes of male and female networks have similar contributions to the overall connectivity, or path length, of their respective intrasexual system, a pattern consistent with the previous analyses taken on fisher network including both sexes (Garroway et al., 2008). In order to confirm these results and validate our models, we evaluated each regression

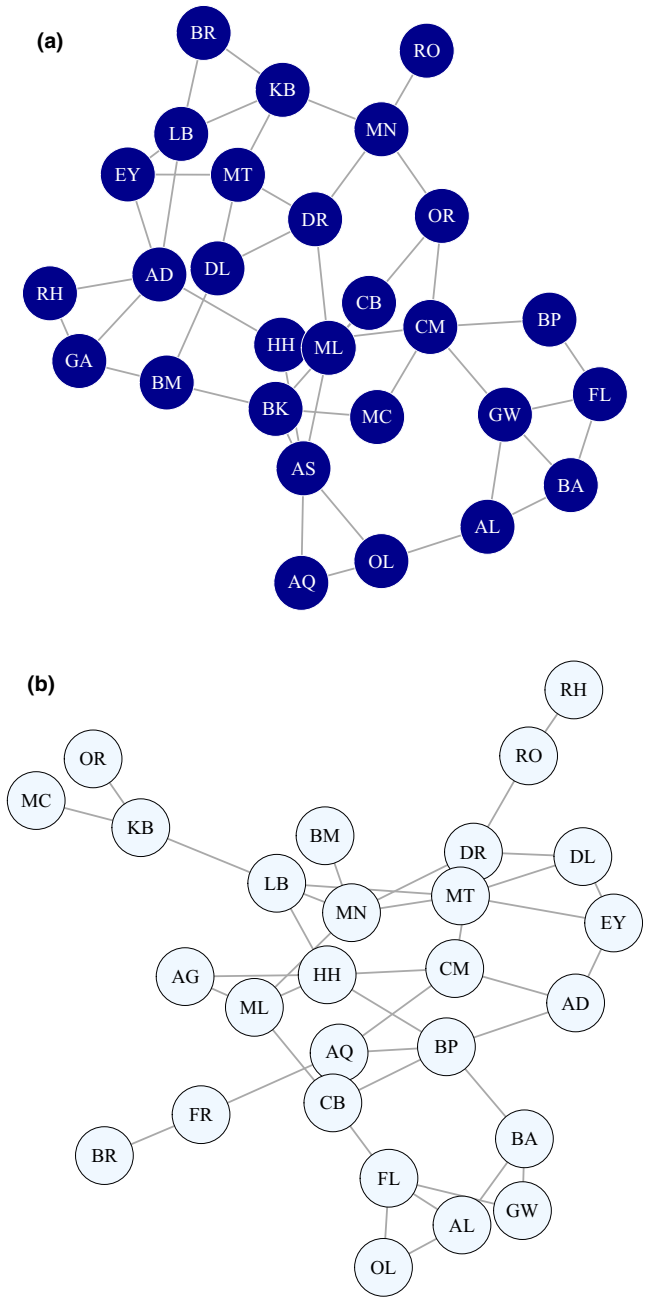

FIGURE 5 Female (a) and male (b) sex-specific graphs. Node sizes are kept constant and edges are mean of significant covariance between the two nodes linked at the nominal level of 0.05

output by the sequential removal of nodes with a low number of individuals (i.e., 2–5, thus removing 1–6 sites). All results were statistically congruent with the selected optimal minimal number of six individuals per site (results not shown).

The male-biased dispersal hypothesis was supported by both randomization tests and topological-based regression models. Congruence between the F_{ST} and relatedness statistics is not surprising considering their close mathematical relationship (Goudet et al., 2002). Significance discrepancies with the other statistics (e.g., F_{IS}, mean and variance of the assignment index) might originate from their underlying sensitivity to the chosen sampling design and intrinsic demographic properties. Evaluating the capacity of each statistic to reliably detect sex-biased dispersal patterns, Goudet et al. (2002) observed that the power of those varies according to population dispersal rate, the magnitude of bias and the number of loci.

TABLE 2 Final beta regression models using sex-specific graphs eigenvector centrality scores as the response variable and demographic variables as uncentered predictors. Slope estimates, standard errors (*SE*) and model's generalized determination coefficient (R^2_{pp}) are given. The link functions used for both location (μ) and precision (ϕ) parameters are also indicated

Response	Predictors	Estimates	*SE*	p^a
Male	logit(μ)			
	Intercept	1.379	0.913	.337
	Prop. Immigrants	−4.437	2.082	.033
	Nb. Individuals	−0.137	0.069	.167
	log(ϕ)			
	Intercept	−0.483	0.848	.748
	Nb. Individuals	0.129	0.078	.322
	R^2_{pp} = 0.358			
Female	logit(μ)			
	Intercept	−0.442	0.742	.780
	Prop. Immigrants	−1.735	1.959	.341
	Nb. Individuals	−0.015	0.062	.778
	log(ϕ)			
	Intercept	0.949	0.246	.586
	R^2_{pp} = 0.030			

[a]Coefficient *p*-values were computed via two-tailed permutation tests along 9,999 iterations.

TABLE 3 Final beta regression models using sex-specific graphs eigenvector centrality scores as the response variable and the landscape variable as a uncentered predictor. Slope estimates, standard errors (*SE*) and model's generalized determination coefficient (R^2_{pp}) are given. The link functions used for both location (μ) and precision (ϕ) parameters are also indicated

Response	Predictors	Estimates	*SE*	p^a
Male	logit(μ)			
	Intercept	0.110	1.315	.941
	Snow Depth	−0.025	0.034	.416
	log(ϕ)			
	Intercept	0.678	0.249	.093
	R^2_{pp} = 0.025			
Female	logit(μ)			
	Intercept	3.208	0.742	.030
	Snow Depth	−0.123	0.020	<.001
	log(ϕ)			
	Intercept	2.675	0.278	<.001
	R^2_{pp} = 0.682			

[a]Coefficient *p*-values were computed via two-tailed permutation tests along 9,999 iterations.

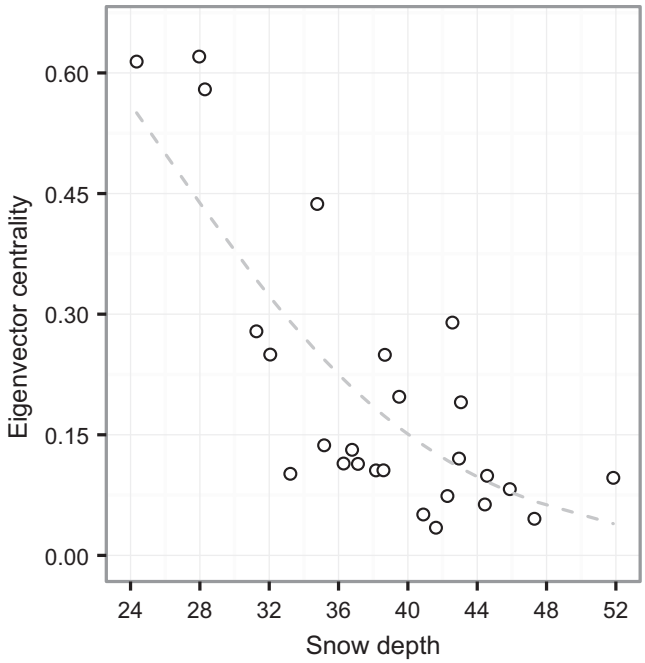

FIGURE 6 Female eigenvector centrality scores as a function of snow depth (cm) with the fitted curve estimated from the corresponding beta regression model (Table 3)

Interestingly, F_{ST}-based tests have been shown to perform well at relatively high dispersal rates (>10%), while vAIc at low (<10%) and mAIc at intermediate levels (Goudet et al., 2002). Important dispersal bias between sexes increases detection probability among statistics but F_{ST} and mAIc shown the greatest power when bias drops, with an increasing capacity more dispersal rate is important. Moreover, when the sampled number of loci is below 20, F_{ST} and mAIc gained power to detect patterns when increasing the number of individual within each population while actually vAIc power eventually decreases. Considering the two sample schemes used in our study (first node-based scheme, $\bar{x}_M$: 10.48 ± 3.32, mean ± *SD*; $\bar{x}_F$: 10.56 ± 3.25; second cluster-based scheme, $\bar{x}_M$: 52.40 ± 14.81; $\bar{x}_F$: 52.80 ± 15.01), number of loci (n = 16) with high level of population structure (Carr, Bowman, Kyle, et al., 2007) and potential high dispersal rate occurring in fisher at high-density condition (Arthur et al., 1993), F_{ST} seems the most powerful statistic to detect potential sex-biased dispersal. In order to evaluate those results, we investigated the sensitivity of the F_{ST} and relatedness statistics across different scenarios of node removal. The two statistics showed high stability across scenarios (see Fig. S1).

Although no difference was detected in centrality distributions between sexes (Mann-Whitney test), regression analyses further confirmed discrepancies between sex-specific graphs by relating topological metrics to demographic data, interestingly suggesting that testing sex-biased dispersal solely through topologies might not be powerful enough to detect differential connectivity patterns. Regressions confirmed that the number of sampled males and females, added here as a control variable, had no effect on sex-specific centrality. Sensitivity analyses investigating the effect of sample size on network topology further supported that averaged centrality scores of both sexes varied weakly with sample sizes in each sex-specific graph (see Fig. S2). The proportion of immigrants, however, described a negative and significant relationship to the male connectivity pattern, supporting the male-biased hypothesis. The observed negative relationship was congruent with previous investigations, a pattern likely driven by density-dependent habitat selection and reciprocating dispersal (Carr, Bowman, & Wilson, 2007; Garroway et al., 2008; Morris et al., 2004). According to these theories, asymmetries in migration directionality occur depending on population trajectories (Morris et al., 2004). Due to density fluctuation in habitat patches, population growth promotes emigration from good to low habitat quality, whereas the inverse happens during population decline. Despite the high-density condition prevailing in the metapopulation and the importance of snow depth in determining the fisher habitat suitability (Carr, Bowman, & Wilson, 2007; Krohn et al., 1995, 2005), we detected no relationship between male connectivity and the snow depth distribution, interestingly suggesting that other unmeasured proximate factors at site might influence the dispersal magnitude of male fishers. Competition for mate availability, for instance, might be an important driver of male dispersal, an idea concordant with the local mate competition hypothesis often supported in species with similar reproductive strategy (Greenwood, 1980; Perrin & Mazalov, 2000). Due to data limitations, however, we cannot preclude the potential effect of local resource competition on male dispersal pattern, which is likely resulting from a mixture of density-dependent effects (Busch, Waser, & Dewoody, 2009).

Male-biased dispersal has been detected in fisher populations of western USA (Aubry & Raley, 2006; Aubry et al., 2004; Matthews et al., 2013). Closer to our study area in Maine, USA, distance bias in fishers was not observed during juvenile dispersal (Arthur et al., 1993) but was observed during breeding dispersal where males underwent extensive travelling, while females tended to remain philopatric (Arthur, Krohn, & Gilbert, 1989). Although females may have the capacity to be as vagile as males, our results showed that only male genetic covariance structure, modelled as a sex-specific graph, was related to dispersal data among genetic clusters. Female genetic connectivity was more restricted to within-cluster (or population) gene flow, which is consistent with the philopatry hypothesis. Fishers are known to be polygynous and to express intrasexual territoriality (Arthur et al., 1993; Powell, 1979) that tends to reduce equality in parental investments (Aubry et al., 2004), and maximize breeding opportunities for territorial males. Nevertheless, as stated above, samples were collected under a colonization event and population growth phase, where positive density-dependent dispersal is suggested (Bowman et al., 2006; Carr, Bowman, & Wilson, 2007). As sex-specific dispersal process has been shown to be variable according to population density level (Busch et al., 2009), we stress that the observed magnitude of bias between male and female patterns should be interpreted in the light of this particular colonization and high-density context. Further investigations are suggested to understand fisher sex-specific dispersal dynamics through varying demographic conditions.

Regressions relating snow depth to node centralities revealed a pronounced deviation between the two sexes, where snow depth has been shown, contrary to our prediction, to reduce genetic connectivity in females while being neutral in males. Differential landscape effects on sex-specific connectivity have been quantified in recent studies (Amos et al., 2014; Elliot et al., 2014; Rioux Paquette et al., 2014; Talbot et al., 2012) and particularly in looking at the effect of snow (Gow & Wiebe, 2014; Rittenhouse, Mong, & Hart, 2015). The proposed node-based approach from intrasexual networks appears to be sensitive to sex-specific functional connectivity and suitable for its analysis. Snow depth has been identified as an important factor affecting fisher dispersal (Aubry & Houston, 1992; Carr, Bowman, & Wilson, 2007; Garroway et al., 2011; Garroway et al., 2008; Krohn et al., 2005; Raine, 1983), and our results suggest that this effect is partitionable by sexes. Sexual dimorphism is considerable in fisher where males are considerably larger and longer (Powell & Leonard, 1983) and demonstrate a significantly greater foot loading than females (Krohn et al., 2005). Beyond this morphological advantage over males, female's sizes have been suggested to respond to selective pressures for lowering energy expenditure under reproductive time (Powell & Leonard, 1983), demonstrating the importance that snow could impose on the daily energy budget. This apparent constraint on females might be related to Arthur et al.'s (1989) observations that male home range locations, also in a high-density condition, were more variable between seasons and years than females, which were more stable. In a more recent study, Lewis, Jenkins, Happe, Manson, and McCalmon (2016) have congruent observations where females were more selective of the core area of their home range than males, selecting specific forested habitat (i.e., mid-sized trees), hypothetically linked to their reproductive and survival needs. Parturition occurs in spring, where the latter third of gestation is taking place under late winter to early spring conditions (Kirkpatrick, 1988; Krohn et al., 2005). The observed reduced functional connectivity and sensitivity to snow depth may reflect tradeoffs that female demonstrate for dispersal and habitat quality and particularly under range expansion, where intrasexual competition for territory due to density dependence effect is likely (Arthur et al., 1993; Carr, Bowman, & Wilson, 2007; Powell, 1979).

In conclusion, this study supports the idea that sex-specific processes can have a differential contribution to the overall genetic structure of wildlife species and can be reflected at different spatial scales and level of analysis (Busch et al., 2009; Roy, Gray, Stoinski, Robbins, & Vigilant, 2014). We found that sex-specific graph topology, using the population graph framework (Dyer & Nason, 2004), was sensitive to intrasexual dispersal dynamics and landscape composition heterogeneity. This site/node-based design is seen as complementary to the distance/edge-based approach most commonly used to date for investigating landscape effects on sex-specific functional connectivity (e.g., Amos et al., 2014; Harrisson, Pavlova, Amos, Radford, & Sunnucks, 2014; Rioux Paquette et al., 2014; Talbot et al., 2012) and we believe that the proposed approach would be applicable to other vagile species where differential sex-specific processes are expected to occur. We suggest a few recommendations (Figure 7) based on our findings: (i) investigators should determine the level of within-node genetic variability and adjust the minimum sample size to generate stable metrics (see Koen et al. (2013) for more information); (ii) topological metrics should be chosen that are appropriate for study objectives, including species autecology (see Newman (2010) or Wasserman and Faust (1994) for theoretical

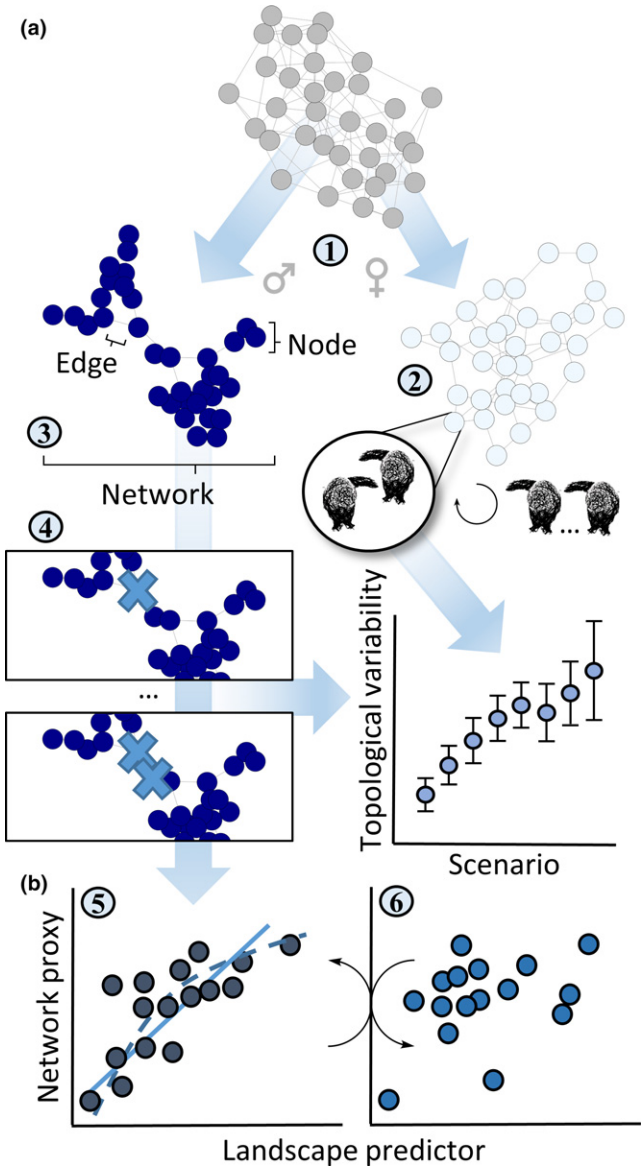

FIGURE 7 A schematic protocol of the different steps followed in this study. The general approach is divided into two main parts: a) data preparation and b) ecological analysis. Step 1 corresponds to the network decomposition, where the population graph is divided into two sex-specific graphs; Step 2 refers to the exploration of within-node genetic variation evaluated through simulation of undersampling scenarios; Step 3 pertains to network proxy selection; Step 4 refers to the evaluation of network measure's resilience to subsampling simulated scenarios; Step 5 is the selection of the adequate modelling approach; and Step 6 is the result validation, comparing outputs from different approaches. Please refer to the Data S2 for more details and a complete protocol along with R codes

aspects and Garroway et al. (2008) for application); (iii) the sensitivity of metrics to varying sampling intensities should be measured using network-, node- or edge-level indices (see Wasserman and Faust (1994) for theoretical aspects and Naujokaitis-Lewis et al. (2013) for application); (iv) adequate modelling approach should be undertaken to assess alternative predictions; and (v) model validation should be undertaken using existing tools (for sex-biased dispersal, e.g., Goudet et al., 2002; Banks & Peakall, 2012). Our results highlight the importance of conducting sex-specific analysis for wildlife conservation by identifying landscape elements (Talbot et al., 2012) or configuration (Amos et al., 2014) that promotes or impedes functional connectivity of wildlife populations, which may remain otherwise cryptic at the population level (Rioux Paquette et al., 2014). Indeed, sex-specific ecological characteristics and selective pressures may lead to a different response of individual behaviour to landscape heterogeneity (Amos et al., 2014; Anadón et al., 2012; Stow et al., 2001). Concordant with previous studies based on large-scale movement behaviour (Elliot et al., 2014) and population genetics analysis (Proctor et al., 2005), our findings underline the necessity of considering demographic groups separately, providing a more accurate conclusion about population landscape permeability and improved conservation strategies.

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DATA ACCESSIBILITY

Individual genotype data are available from the DRYAD digital repository: https://doi.org/10.5061/dryad.167d5. R codes are also publicly available and can be accessed from the GitHub repository: https://github.com/PhilBertrand/SexSpecificGraph.

AUTHOR CONTRIBUTIONS

P.B., J.B. and P.J.W. conceived and performed the study. P.B. analysed the data and wrote the manuscript. J.B., P.J.W., R.J.D. and M.M. commented on the manuscript.

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